

SOLIDWORKS API (.NET VBA C#) — Reference (237/337): IRefPlaneFeatureData- IRefPointFeatureData

SOLIDWORKS API Documentation (converted from CHM)

IRefPlaneFeatureData (Type)

IRefPlaneFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPlaneFeatureData Interface

Allows access to reference plane feature data.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IRefPlaneFeatureData
```

Visual Basic (Usage)

```
Dim instance As IRefPlaneFeatureData
```

C#

```
public interface IRefPlaneFeatureData
```

C++/CLI

```
public interface class IRefPlaneFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData](#).

Example

[Insert Reference Plane \(VBA\)](#)
[Insert Reference Plane \(VB.NET\)](#)
[Insert Reference Plane \(C#\)](#)

Remarks

Before calling any of the methods and properties on this interface, you should determine the reference plane's type because some IRefPlaneFeatureData methods and properties only work with constraint-based reference planes and some only work with non-constrained reference planes. Use [IRefPlaneFeatureData::Type2](#) to determine the type of reference plane.

If IRefPlaneFeatureData::Type2 returns swRefPlaneConstraintBase, then the reference plane is constraint based. To determine if a constraint-based reference plane has angle or offset distances references, call [IRefPlaneFeatureData::Type](#):

- swRefPlaneAngle is returned for angle references.
- swRefPlaneDistance is returned for offset distance references.

Otherwise, the reference plane is not constraint based. Call [IRefPlaneFeatureData::Type](#) to determine its type.

NOTE: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not.

The following table shows which IRefPlaneFeatureData methods and properties to use with which types of reference planes.

Method or property	Constraint-based	Constraint-based with angle or offset distance references	Not constraint-based
AccessSelections	?	?	?
GetSelectionsCount	?	?	?
IAccessSelections	?	?	?
IGetSelections	?	?	?

ISetSelections	X	?	?
ReleaseSelectionAccess	?	?	?
Angle	X	?	?
AngleOrDistance	?	?	X
AutoSize	X	?	?
Constraint	?	?	X
Distance	X	?	?
OriginOnCurve	X	?	?
ProjectionType	X	?	?
Reference	?	?	X
ReverseDirection	X	?	?
Selections	? (get only)	?	?
SolutionIndex	X	?	?
Type	X	?	?
Type2	?	?	X
UseNormalPlane	X	?	?

? = Yes

X = No

Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

Access Diagram

[RefPlaneFeatureData](#)

See Also

[IRefPlaneFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IRefPlane Interface](#)

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IRefPlaneFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPlaneFeatureData Interface

For a list of all members of this type, see [IRefPlaneFeatureData members](#).

Public Methods

Name	Description
------	-------------

	AccessSelections	Gains access to the selections that define a reference plane feature.
	GetSelectionsCount	Gets the number of entities selected to create this reference plane feature.
	IAccessSelections	Gains access to the selections that define a reference plane feature.
	IGetSelections	Gets the selected entities used to create this reference plane feature.
	ISetSelections	Sets the entities to use to create the reference plane feature.
	ReleaseSelectionAccess	Releases access to the selections that created the reference plane feature.

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See Also

[IRefPlaneFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IRefPlane Interface](#)

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IRefPlaneFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPlaneFeatureData Interface

The following tables list the members exposed by [IRefPlaneFeatureData](#).

Public Properties

Name	Description
 Angle	Gets or sets the angle of the reference plane feature.
 AngleOrDistance	Gets or sets the angle or distance of the specified reference for this reference plane feature.
 AutoSize	Gets or sets whether to automatically size the reference plane feature to either the geometry on which it is created or to the bounding box of the model geometry.
 Constraint	Gets or sets the constraint for the specified reference for this reference plane feature.
 Distance	Gets or sets the distance, in meters, to offset the reference plane feature.
 OriginOnCurve	Gets or sets whether to place the origin on the curve for this reference plane feature.
 ProjectionType	Gets or sets the projection type for this on-surface reference plane feature.
 Reference	Gets or sets the reference entity for the specified reference for this reference plane feature.
 ReverseDirection	Obsolete. Superseded by IRefPlaneFeatureData.ReversedReferenceDirection .
 ReversedReferenceDirection	Gets or sets whether to reverse the direction of the specified reference for this reference plane feature.
 Selections	Gets or sets the selected entities used to create the reference plane feature or sets the entities to use to create the reference plane feature.
 SolutionIndex	Gets or sets the intended plane when there are multiple planes from which to select for an on-surface reference plane feature.
 Type	Gets the type of reference plane created in SOLIDWORKS 2009 or earlier. Can also get whether a constraint-based reference plane created in SOLIDWORKS 2010 or has angle or offset distance references. NOTE: This property is a get-only property. Set is not implemented.
 Type2	Gets whether the reference plane is constraint based; thus, created in SOLIDWORKS 2010 and later. NOTE: This property is a get-only property. Set is not implemented.
 UpdatePlane	Gets or sets whether to update this reference plane so that it is parallel to the screen.
 UseNormalPlane	Gets or sets whether to: - Use the plane normal to the selected plane - Automatically size the plane to either the geometry on which it is created or to the bounding box of the model geometry

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Public Methods

Name	Description
------	-------------

 AccessSelections	Gains access to the selections that define a reference plane feature.
 GetSelectionsCount	Gets the number of entities selected to create this reference plane feature.
 IAccessSelections	Gains access to the selections that define a reference plane feature.
 IGetSelections	Gets the selected entities used to create this reference plane feature.
 ISetSelections	Sets the entities to use to create the reference plane feature.
 ReleaseSelectionAccess	Releases access to the selections that created the reference plane feature.

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See Also

[IRefPlaneFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IRefPlane Interface](#)

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IRefPlaneFeatureData Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IRefPlaneFeatureData Interface

For a list of all members of this type, see [IRefPlaneFeatureData members](#).

Public Properties

Name	Description
 Angle	Gets or sets the angle of the reference plane feature.
 AngleOrDistance	Gets or sets the angle or distance of the specified reference for this reference plane feature.
 AutoSize	Gets or sets whether to automatically size the reference plane feature to either the geometry on which it is created or to the bounding box of the model geometry.
 Constraint	Gets or sets the constraint for the specified reference for this reference plane feature.
 Distance	Gets or sets the distance, in meters, to offset the reference plane feature.
 OriginOnCurve	Gets or sets whether to place the origin on the curve for this reference plane feature.
 ProjectionType	Gets or sets the projection type for this on-surface reference plane feature.
 Reference	Gets or sets the reference entity for the specified reference for this reference plane feature.
 ReverseDirection	Obsolete. Superseded by IRefPlaneFeatureData.ReversedReferenceDirection.
 ReversedReferenceDirection	Gets or sets whether to reverse the direction of the specified reference for this reference plane feature.
 Selections	Gets or sets the selected entities used to create the reference plane feature or sets the entities to use to create the reference plane feature.
 SolutionIndex	Gets or sets the intended plane when there are multiple planes from which to select for an on-surface reference plane feature.
 Type	Gets the type of reference plane created in SOLIDWORKS 2009 or earlier. Can also get whether a constraint-based reference plane created in SOLIDWORKS 2010 or has angle or offset distance references. NOTE: This property is a get-only property. Set is not implemented.
 Type2	Gets whether the reference plane is constraint based; thus, created in SOLIDWORKS 2010 and later. NOTE: This property is a get-only property. Set is not implemented.
 UpdatePlane	Gets or sets whether to update this reference plane so that it is parallel to the screen.
 UseNormalPlane	Gets or sets whether to: - Use the plane normal to the selected plane - Automatically size the plane to either the geometry on which it is created or to the bounding box of the model geometry

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See Also

AccessSelections Method (IRefPlaneFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : AccessSelections Method (IRefPlaneFeatureData)

Gains access to the selections that define a reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim TopDoc As System.Object  
Dim Component As System.Object  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    System.object TopDoc,  
    System.object Component  
)
```

C++/CLI

```
System.bool AccessSelections(  
& System.Object^ TopDoc,  
& System.Object^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::AccessSelections](#).

Example

[Modify Plane by Editing Its Definition \(VBA\)](#)

Remarks

Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and the methods and properties belonging to this interface.

To modify a feature in a...

Then...

Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IRefPlaneFeatureData::ReleaseSelectionAccess](#) if you did not

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::IAccessSelections Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

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Angle Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : Angle Property (IRefPlaneFeatureData)

Gets or sets the angle of the reference plane feature.

.NET Syntax

Visual Basic (Declaration)

Property Angle As System.Double

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Double
```

```
instance.Angle = value
```

```
value = instance.Angle
```

C#

```
System.double Angle {get; set;}
```

C++/CLI

```
property System.double Angle {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Angle in radians

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Angle](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::UseNormalPlane Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

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AngleOrDistance Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : AngleOrDistance Property (IRefPlaneFeatureData)

Gets or sets the angle or distance of the specified [reference](#) for this reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property AngleOrDistance( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim Index As System.Integer  
Dim value As System.Double  
  
instance.AngleOrDistance(Index) = value  
  
value = instance.AngleOrDistance(Index)
```

C#

```
System.double AngleOrDistance(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.double AngleOrDistance {  
    System.double get(System.int Index);  
    void set (System.int Index, System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the specified reference as defined in [swRefPlaneReferenceIndex_e](#)

Property Value

Angle or distance

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::AngleOrDistance](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

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AutoSize Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : AutoSize Property (IRefPlaneFeatureData)

Gets or sets whether to automatically size the reference plane feature to either the geometry on which it is created or to the bounding box of the model geometry.

.NET Syntax

Visual Basic (Declaration)

```
Property AutoSize As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Boolean
```

```
instance.AutoSize = value
```

```
value = instance.AutoSize
```

C#

```
System.bool AutoSize {get; set;}
```

C++/CLI

```
property System.bool AutoSize {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically size the reference plane, false to not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::AutoSize](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::UseNormalPlane Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

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Constraint Property (IRefPlaneFeatureData)

[See Also](#)



Gets or sets the constraint for the specified [reference](#) for this reference plane feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property Constraint( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim Index As System.Integer  
Dim value As System.Integer  
  
instance.Constraint(Index) = value  
  
value = instance.Constraint(Index)
```

C#

```
System.int Constraint(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.int Constraint {  
    System.int get(System.int Index);  
    void set (System.int Index, System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the specified [reference](#) as defined in [swRefPlaneReferenceIndex_e](#)

Property Value

Constraint as defined in [swRefPlaneReferenceConstraints_e](#)

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Constraint](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::Reference Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0



Gets or sets the distance, in meters, to offset the reference plane feature.

 .NET Syntax

Visual Basic (Declaration)

Property Distance As System.Double

Visual Basic (Usage) 

```
Dim instance As IRefPlaneFeatureData
Dim value As System.Double

instance.Distance = value

value = instance.Distance
```

C#

```
System.double Distance {get; set;}
```

C++/CLI

```
property System.double Distance {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value
Distance, in radians, to offset the reference plane

 Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Distance](#).

 Example

[Modify Plane by Editing Its Definition \(VBA\)](#)

 Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

 See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

 Availability

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GetSelectionsCount Method (IRefPlaneFeatureData)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : GetSelectionsCount Method (IRefPlaneFeatureData)

Gets the number of entities selected to create this reference plane feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetSelectionsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Integer
```

```
value = instance.GetSelectionsCount()
```

C#

```
System.int GetSelectionsCount()
```

C++/CLI

```
System.int GetSelectionsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of entities selected to create this reference plane feature

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::GetSelectionsCount](#).

Remarks

Call this method before calling [IRefPlaneFeatureData::IGetSelections](#).

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::ISetSelections Method](#)
[IRefPlaneFeatureData::Selections Property](#)
[IRefPlaneFeatureData::ISetSelections Method](#)

Availability

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IAccessSelections Method (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : IAccessSelections Method (IRefPlaneFeatureData)

Gains access to the selections that define a reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections(  
    ByVal TopDoc As ModelDoc2, _  
    ByVal Component As Component2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim TopDoc As ModelDoc2  
Dim Component As Component2
```

```
Dim value As System.Boolean
```

```
value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(  
    ModelDoc2 TopDoc,  
    Component2 Component  
)
```

C++/CLI

```
System.bool IAccessSelections(  
& ModelDoc2^ TopDoc,  
& Component2^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::IAccessSelections](#).

Remarks

Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and the methods and properties belonging to this interface.

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IRefPlaneFeatureData::ReleaseSelectionAccess](#) if you did not

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::AccessSelections Method](#)

Availability

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IGetSelections Method (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : IGetSelections Method (IRefPlaneFeatureData)

Gets the selected entities used to create this reference plane feature.

.NET Syntax

```
Function IGetSelections( _
    ByVal Count As System.Integer _
) As System.Object
```



```
Dim instance As IRefPlaneFeatureData
Dim Count As System.Integer
Dim value As System.Object

value = instance.IGetSelections(Count)
```

```
System.Object IGetSelections(
    System.int Count
)
```

```
System.Object^ IGetSelections(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of selected entities to use to create this reference plane

Return Value

- in-process, unmanaged C++: Pointer to an array of entities of size Count
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IRefPlaneFeatureData::GetSelectionsCount](#) to get the value for Count.

See [Accessing Selections that Define Features](#) for additional details on using this method.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::Selections Property](#)
[IRefPlaneFeatureData::ISetSelections Method](#)

Availability

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ISetSelections Method (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : ISetSelections Method (IRefPlaneFeatureData)

Sets the entities to use to create the reference plane feature.

.NET Syntax

```
Sub ISetSelections( _
    ByVal Count As System.Integer, _
    ByRef Entities As System.Object _
)
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim Count As System.Integer
Dim Entities As System.Object

instance.ISetSelections(Count, Entities)
```

C#

```
void ISetSelections(
    System.int Count,
    ref System.object Entities
)
```

C++/CLI

```
void ISetSelections(
    & System.int Count,
    & System.Object^% Entities
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of selected entities

Entities

- in-process, unmanaged C++: Pointer to an array of entities of size Count

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::GetSelectionsCount Method](#)
[IRefPlaneFeatureData::IGetSelections Method](#)
[IRefPlaneFeatureData::Selections Property](#)

Availability

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OriginOnCurve Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : OriginOnCurve Property (IRefPlaneFeatureData)

Gets or sets whether to place the origin on the curve for this reference plane feature.

.NET Syntax

Visual Basic (Declaration)

Property OriginOnCurve As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Boolean

instance.OriginOnCurve = value

value = instance.OriginOnCurve
```

C#

```
System.bool OriginOnCurve {get; set;}
```

C++/CLI

```
property System.bool OriginOnCurve {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to place the origin on the curve, false to place the origin on the vertex or point

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::OriginOnCurve](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ProjectionType Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : ProjectionType Property (IRefPlaneFeatureData)

Gets or sets the projection type for this on-surface reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ProjectionType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Integer

instance.ProjectionType = value

value = instance.ProjectionType
```

C#

```
System.int ProjectionType {get; set;}
```

C++/CLI

```
property System.int ProjectionType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of a plane on a surface as defined in [swOnSurfacePlaneProjectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::ProjectionType](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Reference Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : Reference Property (IRefPlaneFeatureData)

Gets or sets the reference entity for the specified reference for this reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property Reference( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim Index As System.Integer  
Dim value As System.Object
```

```
instance.Reference(Index) = value
```

```
value = instance.Reference(Index)
```

C#

```
System.object Reference(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.Object^ Reference {
    System.Object^ get(System.int Index);
    void set (System.int Index, System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the specified reference as defined in [swRefPlaneReferenceIndex_e](#)

Property Value

Reference entity

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Reference](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)

[IRefPlaneFeatureData Members](#)

[IRefPlaneFeatureData::ReversedReferenceDirection Property \(\)](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : ReleaseSelectionAccess Method (IRefPlaneFeatureData)

Releases access to the selections that created the reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
```

```
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::ReleaseSelectionAccess](#).

Remarks

[IRefPlaneFeatureData::AccessSelections](#) and [IRefPlaneFeatureData::!AccessSelections](#) put the model in a rollback state to allow access to the selections that define the feature.

Use this method to restore the rollback state if you did not modify the feature, or use [IFeature::ModifyDefinition](#) or [IFeature::!ModifyDefinition2](#) if you did modify the feature.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReverseDirection Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : ReverseDirection Property (IRefPlaneFeatureData)

Obsolete. Superseded by [IRefPlaneFeatureData::ReversedReferenceDirection](#).

.NET Syntax

Visual Basic (Declaration)

Property ReverseDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction, false to not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::ReverseDirection](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReversedReferenceDirection Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : ReversedReferenceDirection Property (IRefPlaneFeatureData)

Gets or sets whether to reverse the direction of the specified reference for this reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ReversedReferenceDirection( _  
    ByVal ReferenceIndex As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim ReferenceIndex As System.Integer  
Dim value As System.Boolean
```

```
instance.ReversedReferenceDirection(ReferenceIndex) = value
```

```
value = instance.ReversedReferenceDirection(ReferenceIndex)
```

C#

```
System.bool ReversedReferenceDirection(  
    System.int ReferenceIndex  
) {get; set;}
```

C++/CLI

```
property System.bool ReversedReferenceDirection {  
    System.bool get(System.int ReferenceIndex);  
    void set (System.int ReferenceIndex, System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReferenceIndex

Index of the reference whose direction to set as defined in [swRefPlaneReferenceIndex_e](#)

Property Value

True to reverse the direction of the specified reference, false to not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::ReversedReferenceDirection](#).

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::Reference Property \(\)](#)

Availability

SOLIDWORKS 2018 SP04, Revision Number 26.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

Selections Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : Selections Property (IRefPlaneFeatureData)

Gets or sets the selected entities used to create the reference plane feature or sets the entities to use to create the reference plane feature.

.NET Syntax

Visual Basic (Declaration)

Property Selections As System.Object

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Object
```

```
instance.Selections = value
```

```
value = instance.Selections
```

C#

```
System.Object Selections {get; set;}
```

C++/CLI

```
property System.Object^ Selections {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of selected entities

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Selections](#).

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::GetSelectionsCount Method](#)
[IRefPlaneFeatureData::GetSelections Method](#)
[IRefPlaneFeatureData::SetSelections Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SolutionIndex Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : SolutionIndex Property (IRefPlaneFeatureData)

Gets or sets the intended plane when there are multiple planes from which to select for an on-surface reference plane feature.

.NET Syntax

Visual Basic (Declaration)

```
Property SolutionIndex As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Integer
```

```
instance.SolutionIndex = value
```

```
value = instance.SolutionIndex
```

C#

```
System.int SolutionIndex {get; set;}
```

C++/CLI

```
property System.int SolutionIndex {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Value indicating intended plane (0-based index)

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::SolutionIndex](#).

Remarks

This property corresponds to the **Other Solutions** check box that appears on the Plane PropertyManager page when there is more than one plane from which to select when creating an on-surface reference plane in SOLIDWORKS.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Type Property (IRefPlaneFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : Type Property (IRefPlaneFeatureData)

Gets the type of reference plane created in SOLIDWORKS 2009 or earlier. Can also get whether a constraint-based reference plane created in SOLIDWORKS 2010 or has angle or offset distance references.

NOTE: This property is a get-only property. Set is not implemented.

Visual Basic Syntax

Visual Basic (Declaration)

Property Type As System.Integer

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Integer

instance.Type = value
value = instance.Type
```

C#

```
System.int Type {get; set;}
```

C++/CLI

```
property System.int Type {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of reference plane as defined in [swRefPlaneType_e](#)

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Type](#).

Example

[Insert Reference Plane \(C#\)](#)
[Insert Reference Plane \(VB.NET\)](#)
[Insert Reference Plane \(VBA\)](#)

Remarks

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlane Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Type2 Property (IRefPlaneFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : Type2 Property (IRefPlaneFeatureData)

Gets whether the reference plane is constraint based; thus, created in SOLIDWORKS 2010 and later.

NOTE: This property is a get-only property. Set is not implemented.

Visual Basic Syntax

```
Property Type2 As System.Integer
```



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Integer
```

```
instance.Type2 = value
```

```
value = instance.Type2
```

```
System.int Type2 {get; set;}
```

```
property System.int Type2 {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of reference plane as defined in [swRefPlaneType_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::Type2](#).

Example

[Insert Reference Plane \(C#\)](#)
[Insert Reference Plane \(VB.NET\)](#)
[Insert Reference Plane \(VBA\)](#)

Remarks

If [IRefPlaneFeatureData::Type2](#) returns [swRefPlaneConstraintBase](#), then the reference plane is constraint based and was created in SOLIDWORKS 2010 or later. To determine if a constraint-based reference plane has angle or offset distances references, call [IRefPlaneFeatureData::Type](#):

- [swRefPlaneAngle](#) is returned for angle references.
- [swRefPlaneDistance](#) is returned for offset distance references.

Otherwise, the reference plane is not constraint based and was created in SOLIDWORKS 2009 or earlier. Call [IRefPlaneFeatureData::Type](#) to determine its type.

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlane Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

UpdatePlane Property (IRefPlaneFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : UpdatePlane Property (IRefPlaneFeatureData)

Gets or sets whether to update this reference plane so that it is parallel to the screen.

.NET Syntax

Visual Basic (Declaration)

Property UpdatePlane As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData
Dim value As System.Boolean

instance.UpdatePlane = value
value = instance.UpdatePlane
```

C#

```
System.bool UpdatePlane {get; set;}
```

C++/CLI

```
property System.bool UpdatePlane {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to update the reference plane so that it is parallel to the screen, false to not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::UpdatePlane](#).

Example

[Update Plane \(C#\)](#)
[Update Plane \(VB.NET\)](#)
[Update Plane \(VBA\)](#)

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UseNormalPlane Property (IRefPlaneFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPlaneFeatureData Interface](#) : UseNormalPlane Property (IRefPlaneFeatureData)

Gets or sets whether to:

- Use the plane normal to the selected plane
- Automatically size the plane to either the geometry on which it is created or to the bounding box of the model geometry

.NET Syntax

Visual Basic (Declaration)

Property UseNormalPlane As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRefPlaneFeatureData  
Dim value As System.Boolean
```

```
instance.UseNormalPlane = value
```

```
value = instance.UseNormalPlane
```

C#

```
System.bool UseNormalPlane {get; set;}
```

C++/CLI

```
property System.bool UseNormalPlane {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use a plane normal to the selected plane and automatically size the plane, false to not

Visual Basic for Applications (VBA) Syntax

See [RefPlaneFeatureData::UseNormalPlane](#).

Remarks

If this property is true, then you can get or set [IRefPlaneFeatureData::Angle](#).

IMPORTANT: Reference planes created in SOLIDWORKS 2010 or later are constraint based; reference planes created in SOLIDWORKS 2009 or earlier are not. See the **Remarks** section in the [IRefPlaneFeatureData interface](#) topic for details about reference planes and this interface.

See Also

[IRefPlaneFeatureData Interface](#)
[IRefPlaneFeatureData Members](#)
[IRefPlaneFeatureData::AutoSize Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

IRefPoint (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IRefPoint Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPoint Interface

Allows access to reference points.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IRefPoint
```

Visual Basic (Usage)



```
Dim instance As IRefPoint
```

C#

```
public interface IRefPoint
```

C++/CLI

```
public interface class IRefPoint
```

Visual Basic for Applications (VBA) Syntax

See [RefPoint](#).

Example

[Insert Reference Points \(VBA\)](#)
[Get Reference Point Data \(VBA\)](#)
[Get Reference Point Data \(VB.NET\)](#)
[Get Reference Point Data \(C#\)](#)

Accessors

[ICoincidentMateFeatureData::EntitiesToMate](#)
[IConcentricMateFeatureData::EntitiesToMate](#)
[IFeature::GetSpecificFeature2](#)
[IHingeMateFeatureData::EntitiesToMate](#)
[IPartialEdgeFilletData::ReferenceOffsetEnd](#)
[IPartialEdgeFilletData::ReferenceOffsetStart](#)
[IPrimaryMemberPointLengthFeatureData::GetPoints](#)
[ISMGussetFeatureData::ReferencePoint](#)
[IThreadFeatureData::ProfileLocation](#)
[ITwoMemberCornerTreatmentFeatureData::MiterTrimPlanePoint](#)
[IUniversalJointMateFeatureData::JointPoint](#)

Access Diagram

[RefPoint](#)

See Also

[IRefPoint Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IRefPoint Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IRefPoint Interface

For a list of all members of this type, see [IRefPoint members](#).

Public Methods

Name	Description
 GetRefPoint	Gets the parameters for this reference point.

[Top](#)

See Also

[IRefPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::InsertRefPoint Method](#)
[IRefPointFeatureData Interface](#)

IRefPoint Interface Members

[See Also](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IRefPoint Interface

The following tables list the members exposed by [IRefPoint](#).

Public Methods

Name	Description
 GetRefPoint	Gets the parameters for this reference point.

[Top](#)

See Also

[IRefPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::InsertRefPoint Method](#)
[IRefPointFeatureData Interface](#)

GetRefPoint Method (IRefPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPoint Interface](#) : GetRefPoint Method (IRefPoint)

Gets the parameters for this reference point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRefPoint() As MathPoint
```

Visual Basic (Usage)

```
Dim instance As IRefPoint  
Dim value As MathPoint  
  
value = instance.GetRefPoint()
```

C#

```
MathPoint GetRefPoint()
```

C++/CLI

```
MathPoint^ GetRefPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Parameters for this reference point](#)

Visual Basic for Applications (VBA) Syntax

See [RefPoint::GetRefPoint](#).

Example

[Get Reference Point Data \(VBA\)](#)
[Insert Reference Points \(VBA\)](#)

See Also

[IRefPoint Interface](#)
[IRefPoint Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

IRefPointFeatureData (Type)

IRefPointFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPointFeatureData Interface

Allows access to reference point feature data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IRefPointFeatureData
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData
```

C#

```
public interface IRefPointFeatureData
```

C++/CLI

```
public interface class IRefPointFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData](#).

Example

[Get Reference Point Data \(VBA\)](#)
[Get Reference Point Data \(VB.NET\)](#)
[Get Reference Point Data \(C#\)](#)

Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

Access Diagram

[RefPointFeatureData](#)

See Also

[IRefPointFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IRefPoint Interface](#)

IRefPointFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPointFeatureData Interface

The following tables list the members exposed by [IRefPointFeatureData](#).

Public Properties

Name	Description
 AlongCurveOption	Gets or sets whether to enable or disable the option to create the reference point along a curve or to create multiple reference points.

 Distance	Gets the distance at which the reference point was created or sets the distance at which to create the reference point.
 Selections	Gets the entities selected to create the reference point or sets the entities to create the reference point.
 Type	Gets or sets the type of reference point.

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Public Methods

Name	Description
 AccessSelections	Accesses the selections used to create the reference point feature.
 GetSelectionsCount	Gets the number of entities selected to use to create the reference point.
 IAccessSelections	Accesses the selections used to create the reference point feature.
 IGetSelections	Gets the selected entities to use to create the reference points.
 ISetSelections	Sets the number of entities to use to create the reference points.
 ReleaseSelectionAccess	Releases access to the selections that created this reference point feature.

[Top](#)

See Also

[IRefPointFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IRefPoint Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRefPointFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPointFeatureData Interface

For a list of all members of this type, see [IRefPointFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Accesses the selections used to create the reference point feature.
 GetSelectionsCount	Gets the number of entities selected to use to create the reference point.
 IAccessSelections	Accesses the selections used to create the reference point feature.
 IGetSelections	Gets the selected entities to use to create the reference points.
 ISetSelections	Sets the number of entities to use to create the reference points.
 ReleaseSelectionAccess	Releases access to the selections that created this reference point feature.

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See Also

[IRefPointFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IRefPoint Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRefPointFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRefPointFeatureData Interface

For a list of all members of this type, see [IRefPointFeatureData members](#).

Public Properties

Name	Description
------	-------------

 AlongCurveOption	Gets or sets whether to enable or disable the option to create the reference point along a curve or to create multiple reference points.
 Distance	Gets the distance at which the reference point was created or sets the distance at which to create the reference point.
 Selections	Gets the entities selected to create the reference point or sets the entities to create the reference point.
 Type	Gets or sets the type of reference point.

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See Also

[IRefPointFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IFeatureManager::InsertReferencePoint Method](#)
[IRefPoint Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AccessSelections Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : AccessSelections Method (IRefPointFeatureData)

Accesses the selections used to create the reference point feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _
    ByVal TopDoc As System.Object, _
    ByVal Component As System.Object _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IRefPointFeatureData
Dim TopDoc As System.Object
Dim Component As System.Object
Dim value As System.Boolean

value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::AccessSelections](#).

Example

See [IRefPointFeatureData](#) examples.

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is <code>NULL</code>
Assembly	- TopDoc is the <code>IModelDoc2</code> object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IRefPointFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)
[IRefPointFeatureData::IAccessSelections Method](#)

Availability

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SOLIDWORKS API Help

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AlongCurveOption Property (IRefPointFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : AlongCurveOption Property (IRefPointFeatureData)

Gets or sets whether to enable or disable the option to create the reference point along a curve or to create multiple reference points.

.NET Syntax

Visual Basic (Declaration)

```
Property AlongCurveOption As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim value As System.Integer
```

```
instance.AlongCurveOption = value
```

```
value = instance.AlongCurveOption
```

C#

```
System.int AlongCurveOption {get; set;}
```

C++/CLI

```
property System.int AlongCurveOption {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True enables this option, false does not

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::AlongCurveOption](#).

Example

See [IRefPointFeatureData](#) examples.

Example

[Insert Reference Points \(VBA\)](#)

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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Distance Property (IRefPointFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : Distance Property (IRefPointFeatureData)

Gets the distance at which the reference point was created or sets the distance at which to create the reference point.

.NET Syntax

Visual Basic (Declaration)

Property Distance As System.Double

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim value As System.Double
```

```
instance.Distance = value
```

```
value = instance.Distance
```

C#

```
System.double Distance {get; set;}
```

C++/CLI

```
property System.double Distance {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData.Distance](#).

Example

See [IRefPointFeatureData](#) examples.

Example

[Insert Reference Points \(VBA\)](#)

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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GetSelectionsCount Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : GetSelectionsCount Method (IRefPointFeatureData)

Gets the number of entities selected to use to create the reference point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSelectionsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim value As System.Integer  
  
value = instance.GetSelectionsCount()
```

C#

```
System.int GetSelectionsCount()
```

C++/CLI

```
System.int GetSelectionsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of selected entities

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::GetSelectionsCount](#).

Remarks

Call this method before calling [IRefPointFeatureData::IGetSelections](#).

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)
[IRefPointFeatureData::IGetSelections Method](#)
[IRefPointFeatureData::Selections Property](#)

Availability

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IAccessSelections Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : IAccessSelections Method (IRefPointFeatureData)

Accesses the selections used to create the reference point feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IAccessSelections( _  
    ByVal TopDoc As ModelDoc2, _  
    ByVal Component As Component2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim TopDoc As ModelDoc2  
Dim Component As Component2  
Dim value As System.Boolean  
  
value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(  
    ModelDoc2 TopDoc,  
    Component2 Component  
)
```

C++/CLI

```
System.bool IAccessSelections(  
& ModelDoc2^ TopDoc,  
& Component2^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::IAccessSelections](#).

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you modified the feature
- [IRefPointFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)
[IRefPointFeatureData::AccessSelections Method](#)

Availability

IGetSelections Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : IGetSelections Method (IRefPointFeatureData)

Gets the selected entities to use to create the reference points.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSelections( _  
    ByVal Count As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim Count As System.Integer  
Dim value As System.Object  
  
value = instance.IGetSelections(Count)
```

C#

```
System.object IGetSelections(  
    System.int Count  
)
```

C++/CLI

```
System.Object^ IGetSelections(  
&    System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of selected entities

Return Value

- in-process, unmanaged C++: Pointer to an array of selected entities of size Count
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::IGetSelections](#).

Remarks

Call [IRefPointFeatureData::GetSelectionsCount](#) before calling this method to get the value for Count.

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)
[IRefPointFeatureData::ISetSelections Method](#)
[IRefPointFeatureData::Selections Property](#)

Availability

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ISetSelections Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : ISetSelections Method (IRefPointFeatureData)

Sets the number of entities to use to create the reference points.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSelections( _  
    ByVal Count As System.Integer, _  
    ByRef Entities As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim Count As System.Integer  
Dim Entities As System.Object  
  
instance.ISetSelections(Count, Entities)
```

C#

```
void ISetSelections(  
    System.int Count,  
    ref System.object Entities  
)
```

C++/CLI

```
void ISetSelections(  
& System.int Count,  
& System.Object^% Entities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of selected entities

Entities

- in-process, unmanaged C++: Pointer to an array of entities of size Count
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [in-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[IRefPointFeatureData Interface](#)

[IRefPointFeatureData Members](#)

[IRefPointFeatureData::GetSelectionsCount Method](#)

[IRefPointFeatureData::IGetSelections Method](#)

[IRefPointFeatureData::Selections Property](#)

Availability

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ReleaseSelectionAccess Method (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : ReleaseSelectionAccess Method (IRefPointFeatureData)

Releases access to the selections that created this reference point feature.

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)

```
Dim instance As IRefPointFeatureData  
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::ReleaseSelectionAccess](#).

Example

See [IRefPointFeatureData](#) examples.

Remarks

[IRefPointFeatureData::AccessSelections](#) and [IRefPointFeatureData::!AccessSelections](#) put the model in a rollback state to allow access to the selections that define the feature.

Use this method to restore the rollback state if you did not modify the feature, or use [IFeature::ModifyDefinition](#) or [IFeature::!ModifyDefinition2](#) if you did modify the feature.

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)

Availability

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Selections Property (IRefPointFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : Selections Property (IRefPointFeatureData)

Gets the entities selected to create the reference point or sets the entities to create the reference point.

Visual Basic (Declaration)

```
Property Selections As System.Object
```

Visual Basic (Usage)

```
Dim instance As IRefPointFeatureData  
Dim value As System.Object
```

```
instance.Selections = value  
value = instance.Selections
```

C#

```
System.Object Selections {get; set;}
```

C++/CLI

```
property System.Object^ Selections {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of selected entities

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::Selections](#).

Example

See [IRefPointFeatureData](#) examples.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)
[IRefPointFeatureData::GetSelectionsCount Method](#)
[IRefPointFeatureData::iGetSelections Method](#)
[IRefPointFeatureData::iSetSelections Method](#)

Availability

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Type Property (IRefPointFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRefPointFeatureData Interface](#) : Type Property (IRefPointFeatureData)

Gets or sets the type of reference point.

.NET Syntax

Visual Basic (Declaration)

```
Property Type As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRefPointFeatureData  
Dim value As System.Integer
```

```
instance.Type = value
```

```
value = instance.Type
```

C#

```
System.int Type {get; set;}
```

C++/CLI

```
property System.int Type {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of reference point as defined by [swRefPointType_e](#)

Visual Basic for Applications (VBA) Syntax

See [RefPointFeatureData::Type](#).

Example

See [IRefPointFeatureData](#) examples.

Example

[Insert Reference Points \(VBA\)](#)

See Also

[IRefPointFeatureData Interface](#)
[IRefPointFeatureData Members](#)

Availability

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